

EXOhSPEC V19: 5-hour passive thermal investigation

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Aim. Performed a 5-hour monitor-only run to investigate the origin of the repeatable thermal oscillation observed around the camera system. The main question is why a strong air-conditioning-related oscillation appears in the cooling and camera temperatures while the measured internal instrument environment remains comparatively stable.

Dataset. The run contains **301 science frames over 5.00 h**, together with **3612 LK220 measurements** sampled at approximately 5 s cadence. The bench TEC remained externally fixed at 22 °C and no AO correction was applied during this passive run.

Environmental and camera temperatures. The bench TEC remained very stable at **22.001 ± 0.012 °C**. The internal ECU environmental temperature was **23.586 ± 0.008 °C**, with only **0.034 °C peak-to-peak** variation.

The BME sensor provides the general HVAC/environmental measurement. Its temperature was **23.623 ± 0.145 °C**; however, this standard deviation includes the initial settling from 24.81 °C. The separate PT104 probe positioned toward the air conditioner measured **22.642 ± 0.141 °C** and showed a clear **0.535 °C peak-to-peak periodic oscillation**.

The three camera-related PT104 probes independently measured:

Probe	Mean [°C]	σ [°C]	Peak-to-peak [°C]
Camera left	24.928	0.104	0.396
Camera right	24.450	0.075	0.314
Camera top	25.310	0.133	0.530

Relative to the internal ECU temperature, the camera left, right and top were hotter by approximately **+1.34, +0.86 and +1.72 °C**, respectively. Relative to the PT104 air-conditioner probe, these temperature differences were approximately **+2.29, +1.81 and +2.67 °C**.

This is an important result: the camera is locally hotter than the measured surrounding environment and its thermal oscillation is much larger than the variation measured by the internal ECU sensor. Therefore, uniform heating and cooling of the air inside the enclosure cannot by itself explain the observed camera behaviour.

Cooling-system behaviour. The LK220 was held at a nominal setpoint of 22.0 °C. Its measured TACT was **21.993 ± 0.149 °C**, with **0.490 °C peak-to-peak** variation. The LK220 TEC current changed strongly between approximately **2.1 and 5.4 A (4.00 ± 0.93 A)**. The camera CoolerPower was **72.58 ± 0.55%**, mainly switching between 72–73% with brief excursions to 74%.

The correlation results show that the camera temperatures are much more closely associated with the cooling system than with the measured air temperature. LK220 TACT correlates with camera left, right and top at $r = 0.896, 0.902$ and 0.907 , respectively. Camera CoolerPower gives $r = 0.873, 0.842$ and 0.854 . In comparison, the PT104 air-conditioner probe gives $r = 0.345, 0.449$ and 0.459 , while BME/HVAC temperature gives only $r = 0.089, 0.067$ and 0.143 .

A strong common period of approximately **33.4 minutes** is present in the PT104 air-conditioner signal, LK220 TACT, LK220 TEC current, camera temperatures, CoolerPower, OPL and dY .

My present working interpretation is therefore: air-conditioning oscillation → LK220 thermal response → coolant / camera heat sink → camera cooler response → camera-body temperature → optical/spectral drift.

The air-conditioning cycle may be the initiating periodic disturbance, but the data suggest that it is transferred into the camera primarily through the **cooling system rather than through uniform heating of the enclosure air**. The LK220 coolant-control loop and the camera's own TEC control loop are also thermally connected. Their different thermal time constants may allow a relatively modest external disturbance to produce a much clearer oscillation at the camera.

The spectral response is dominated by the Y direction: $dX_{\text{RMS}} = 0.076$ **px** and $dY_{\text{RMS}} = 0.537$ **px**. Only **54.2%** of the measured dY positions remained inside ± 0.5 px.

This does not yet prove that the LK220 itself is faulty. Following Bill's suggestion, one PT104 probe should be placed directly onto the camera heat sink beside the two coolant connections. The critical test is whether the measured sequence becomes **LK220 TACT → camera heat sink → camera left/right/top → OPL/ dY** .

The unresolved physical question is therefore: **why is an approximately 0.5 °C air-conditioning oscillation transferred so strongly into the cooling/camera system while the internal ECU temperature changes by only 0.034 °C peak-to-peak, and why does the camera remain locally 1–2 °C hotter than the measured surrounding environment?**

Full-run diagnostic - some plots

EXOhSPEC V19 — 5 h thermal / chiller / spectrograph diagnostics

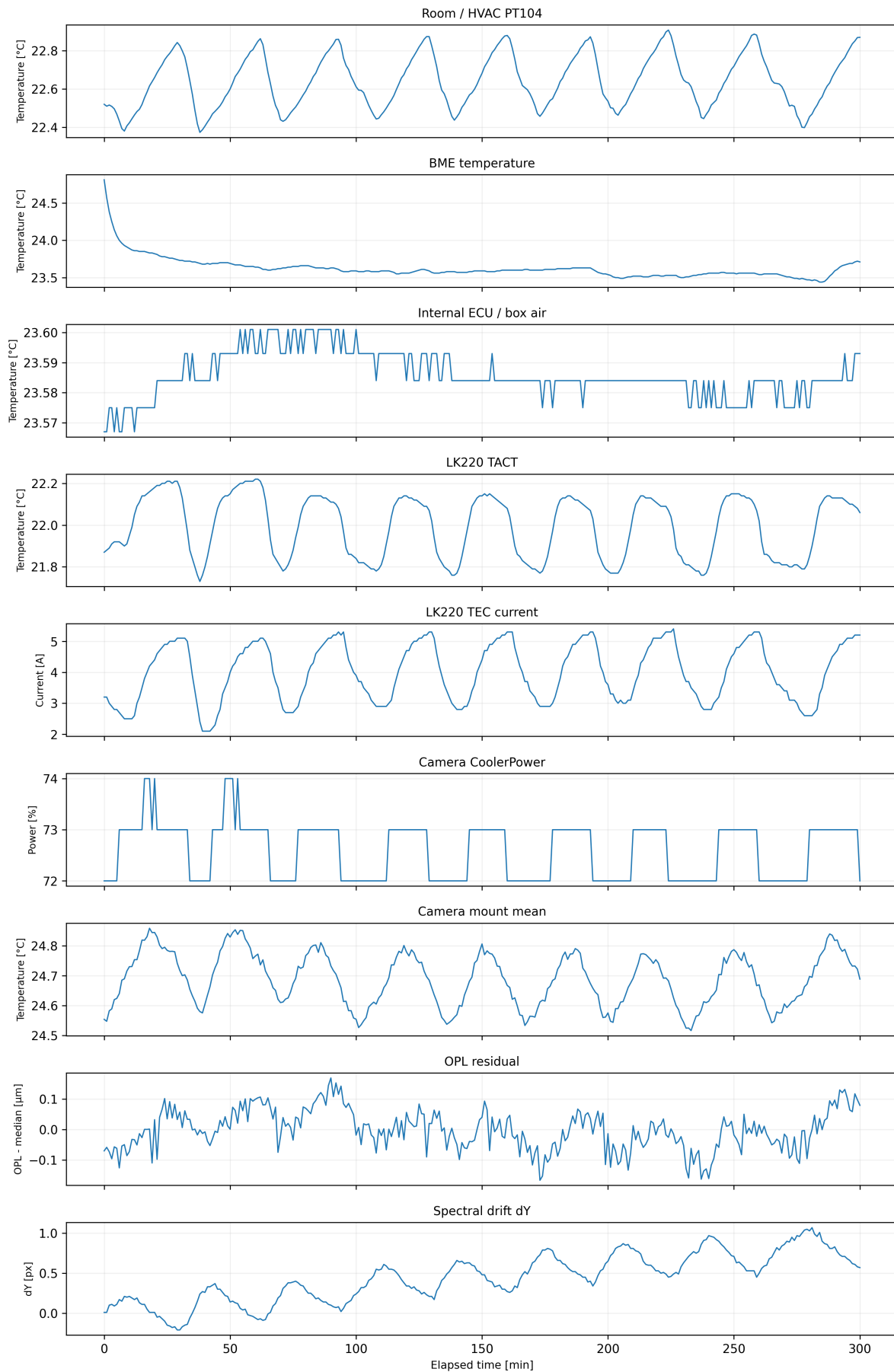


Figure 1. Nine-channel full-run comparison of the environmental, cooling-system, camera, metrology and spectral measurements. The repeated cycle is visible in the PT104 air-conditioner probe, LK220 TACT/current, CoolerPower, camera temperatures and dY , whereas the internal ECU temperature remains comparatively stable.

Cooling-chain evidence and quantitative comparison

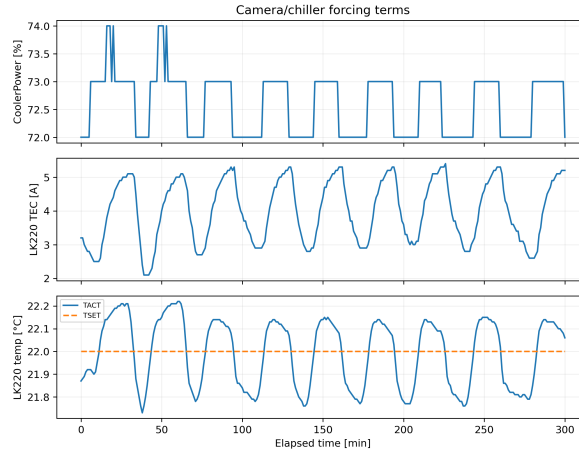


Figure 2. Camera CoolerPower, LK220 TEC current and LK220 TACT. The LK220 current changes from approximately 2.1–5.4 A while TACT oscillates about the nominal 22 °C setpoint.

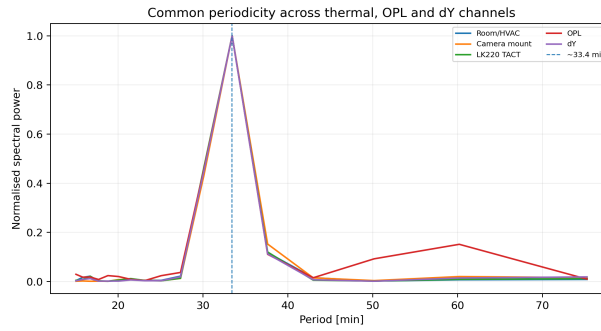


Figure 3. Detrended periodicity comparison showing a dominant common component near 33.4 min in the PT104 air-conditioner probe, LK220/camera variables, OPL and dY .

Thermal statistics and correlation with individual camera probes

Source / channel	Mean	σ	P-P	r_{Left}	r_{Right}	r_{Top}
Bench TEC	22.001 °C	0.012	0.073	—	—	—
Internal ECU	23.586 °C	0.008	0.034	0.104	0.154	0.119
BME/HVAC T	23.623 °C	0.145	1.370*	0.089	0.067	0.143
PT104 AC probe	22.642 °C	0.141	0.535	0.345	0.449	0.459
Camera left	24.928 °C	0.104	0.396	—	—	—
Camera right	24.450 °C	0.075	0.314	—	—	—
Camera top	25.310 °C	0.133	0.530	—	—	—
LK220 TACT	21.993 °C	0.149	0.490	0.896	0.902	0.907
LK220 TEC current	4.00 A	0.93	3.30	0.367	0.474	0.504
CoolerPower	72.58%	0.55	2.00	0.873	0.842	0.854

*BME peak-to-peak includes the initial thermal settling period.

Camera temperature excess and spectral stability

Camera probe	ΔT_{ECU}	$\Delta T_{PT104 AC}$	Spectral metric	Result
Left	+1.34 °C	+2.29 °C	dX_{RMS}	0.076 px
Right	+0.86 °C	+1.81 °C	dY_{RMS}	0.537 px
Top	+1.72 °C	+2.67 °C	dY within ± 0.3 px	34.2%
			dY within ± 0.5 px	54.2%
			OPL vs dY	$r = -0.45$

Main quantitative result. All three camera probes correlate much more strongly with LK220 TACT ($r = 0.896$ – 0.907) and camera CoolerPower ($r = 0.842$ – 0.873) than with either BME/HVAC temperature or the internal ECU temperature. The internal ECU varies by only 0.034 °C peak-to-peak, while the PT104 air-conditioner probe, LK220 TACT and camera probes vary by approximately 0.3–0.5 °C. The camera top is also approximately 1.72 °C hotter than the internal ECU measurement.

Working physical interpretation. The current evidence is most consistent with the sequence

air-conditioning oscillation → LK220 thermal response → coolant / camera heat sink → camera thermal response → OPL/ dY

rather than direct uniform heating of the camera by the air inside the box.

Next decisive measurement. Following Bill's suggestion, one PT104 probe should be placed directly on the camera heat sink beside the two coolant connections while retaining the remaining camera probes, PT104 AC reference, BME/HVAC measurement, LK220 telemetry, CoolerPower, OPL and spectral-position measurements. Basically what we can do here is the current PT104 probe which is towards ac can go to that part of the camera which Bill mentioned and the BME which is hanging on ceiling can be kept near AC instead for monitoring. WE NEED MORE TEMPERATURE PROBESSSSS XD INVESTIGATION IN PROGRESS....
The critical diagnostic is whether the time sequence becomes

$$T_{LK220} \rightarrow T_{\text{heat sink}} \rightarrow T_{\text{camera}} \rightarrow OPL/dY.$$

This would determine whether the missing thermal path is located directly at the coolant–camera interface.